07. AI Fluency

AI Fluency

↳ Ability to use AI effectively, critically, and responsibly.
know when, how, and where to use AI + verify its output.

AI as Coach v/s Teammate

Coach → helps you learn / improve (learn)
Eg: explain concept, give hints, review solution.
Teammates → helps you perform work (work)
eg: generate code, write tests, debug, document.

Core AI fluency Skills

→ Task documentation → clear decomposition → content provision
→ Iterative interaction → output evaluation → verification
→ Problem solving → Tool selection → Responsible AI use.

Task Decomposition

Break a large task into smaller, manageable steps.
Big Task → Smaller tasks → Execute → Test → Review.
Benefits: easier to understand, verify; test; correct.

Good AI Request

Goal + Context + constraints + Expected Output *
eg: Goal ⇒ Explain Recursion ~~Context~~ Constraint: Simple English
Context: Beginner Output: 5 points + example.

Iterative Interaction

- Don't expect the first response to be perfect.
Ask → Review → Identify Problem → Refine → Ask Again → Verify. *

Output Verification

→ Never blindly trust AI output. *
→ check: correctness, sources / claims, edge cases, requirements, current information, code by running / testing it.

Hallucination

- AI may produce information that sounds correct but is false / unsupported.
Response: Ask for sources, Restrict to provided information, and verify independently.

AI limitations

→ Incorrect / outdated information → Missing content.
→ Ambiguity → Hallucinations
→ Reasing errors → Bias
→ security / Privacy Risks → Incorrect code.

Choose Right AI Capability

Simple task → LLM
Current / external info → Retrieval / Tool
Company-specific info → Content / RAG.
Multistep action → Agent.

AI + Critical Thinking

AI generates → you Question → you verify → you decide.
Ask: is it correct? what assumptions? Reliable source?
Meets requirement? what can go wrong?

AI for Software Development

AI can help with: Requirements → Code → Debugging → Refactoring → Testing → Documentation → Review.
But generated code must be: Understood + Reviewed + Tested.

AI for learning

AI can act as: Teacher, Tutor, Reviewer, Interviewer, Debugging partner, Practice Generator.

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Responsible AI

Privacy → protect sensitive data
Security → don't blindly execute AI output
IP → understand content usage
Human oversight → review important decisions.
Transparency → disclose AI assistance when appropriate.

Human-in-the-loop

for imp / high impact actions: AI recommendation → Human Review → Approval → Action.

Imp Points

- AI fluency = effective + critical + responsible AI use.
- AI output ≠ automatically truth.
- Human responsible for verificatn & imp. decisions.
- AI should augment human capability, not replace understandg